An Interactive Dashboard for Analyzing Food Prices in Nigeria, 2007 to 2026

Abstract

This project turned a raw spreadsheet of monthly food prices from 66 Nigerian markets (January 2007 to August 2026) into an interactive Power BI dashboard. Before building any chart, I explored the data, found problems that would have produced wrong results (summary rows mixed in with real markets, uneven market coverage, unclear units), and decided how to handle each one. I then re-organized the data into a clean model in Excel and Power BI, wrote the calculations as measures, and built a three-page dashboard covering the overall food price index, individual food prices, inflation, and differences between states. The analysis shows that average rice prices rose about 17 times between January 2007 and August 2026, that the food price index nearly doubled between 2023 and 2024, and that rice prices in 2025 differed by about 49 percent between the highest-priced and lowest-priced states. The main limitations are that most markets are in the North East and that the price units are not documented.

1. Introduction

1.1 Background

Food is one of the largest items in a Nigerian household budget, so changes in food prices affect families directly. Monitoring these changes across time and place helps explain where and when pressure on households is greatest. The dataset I used records closing, high and low prices for eight foods, plus year-on-year inflation and an overall food price index, for many markets.

1.2 Objectives

1. Understand the structure, quality and limits of the dataset before analyzing it.
2. Clean and organize the data into a model that gives correct results.
3. Build a dashboard that answers four questions: what is happening to food prices overall, how much has each food cost over time, how fast are prices rising, and where are prices highest and lowest.
4. Document the decisions made so that the work can be understood and repeated.

2. The Data

2.1 Structure

Item	Detail
Size	17,464 rows and 43 columns
Period	January 2007 to August 2026 (236 months)
One row means	One location in one month

Item	Detail
Foods	Rice, beans, yam, gari, fish, eggs, beef, milk
Source	[State the original source here once confirmed]

Each food has four columns: the closing price (c_), the monthly high (h_), the monthly low (l_) and year-on-year inflation (inflation_). There are also two composite columns, the food price index and its inflation. Location is described by state, local government area (LGA), market name, a geo_id code and latitude and longitude. The file does not state the unit of the prices (likely Naira per kilogram or per standard unit), which I treat as an open question.

2.2 The location columns

Places are organized as State, then LGA, then Market. The real markets cover 14 states, 47 LGAs and 66 markets. The file also contains 8 summary rows that sit in the same columns: five Geopolitical Zone rows, a National Average row, a Market Average row and a row labelled Kano State (1). These are not markets, and including them in averages would double-count. Market Average, for example, has the same label in the state, LGA and market columns and has no coordinates, so it is neither a state nor an LGA.

3. Data Exploration and Cleaning

3.1 Exploring the data

I first profiled the whole file: the number of rows and columns, the date range, the number of markets per state, the price ranges of each food, the inflation columns, and the missing values. I confirmed that the data is a balanced panel (every location has exactly 236 months, with no duplicate months and no zero or negative prices) and that in every row the closing price lies between the monthly low and high. I also checked how the inflation column was calculated by testing it against one market's prices, and it matched a 12-month percentage change exactly.

3.2 Data quality issues and my decisions

Issue	What I found	Decision
Summary rows mixed with markets	8 rows are not real markets	Added a location_type column (Market, Zone, Benchmark, Unverified) and counted Market rows only
Kano State (1)	A separate series with its own coordinates; its values differ from the only other Kano market (Dawanau) but move with it (correlation 0.995)	Marked Unverified and excluded from rankings until the source explains it. I did not relabel it as a zone because the label would no longer describe

Issue	What I found	Decision
		the data
geo_id lengths differ	The ID is built from the coordinates, so its length varies with the number of digits	No change needed; kept as text
Uneven coverage	Borno has 28 markets and Yobe 18, while nine states have one each. The North East holds 54 of 66 markets	Compared at market level, used medians for inflation, and stated the skew on the dashboard
Blank inflation in 2007	A yearly change needs a prior year	Left blank, because a missing value is not zero
Extreme inflation values	Up to about 984 percent, mostly from small, volatile markets	Kept the data, used the median, and set chart axis ranges by hand
Unknown units and nominal prices	Not documented; prices are not adjusted for inflation	Flagged for confirmation; used multiples and indexes for long-run comparison
No region column	Not in the file	Created a Region column myself, to be verified against an official source

I also found that the National Average and Market Average rows cannot be reproduced from the market rows, so I treated them as separate benchmarks rather than calculating with them.

3.3 The cleaning process, step by step

I followed one rule throughout: flag problems instead of deleting data, and work only on copies. No row of the raw file was deleted or changed, so every decision can be traced and reversed.

1. **Protected the original.** I saved a working copy and built every later table from copies of the raw sheet.
2. **Checked the data types.** I confirmed that price_date was stored as a real date (a text date would not link to the Date table), that all price and inflation columns were numbers, and that geo_id was text.
3. **Checked for duplicates.** I built a PivotTable of market against the count of months. Every location had exactly 236 rows, so there were no duplicate market-months and no rows to remove.
4. **Checked for missing values.** The price columns had no blanks. Each inflation column had 888 blanks (74 locations times the 12 months of 2007), and only the Market Average row lacked coordinates. I left the blanks as they were, because a missing value is not the same as zero and filling it in would invent data.

5. **Checked that the numbers make sense.** I confirmed there were no zero or negative prices and that, for all eight foods, the monthly low was never above the close and the close was never above the high.
6. **Identified the rows that are not markets.** I found the 8 summary rows (section 2.2) and flagged them with a location_type column, rather than deleting them.
7. **Rebuilt the location columns.** The original state column mixed real states with labels such as Market Average, so I created a State column that holds a state only for market rows and n/a otherwise. I added a Region column from a lookup sheet and gave the columns readable names.
8. **Handled the uncertain series.** I marked Kano State (1) as Unverified, wrote the reason in a notes column, and excluded it from rankings.
9. **Left geo_id alone.** Its different lengths come from the coordinates it is built from, and every ID is unique, so it works as a key as it is.
10. **Created the date fields.** I generated a separate Date table with one row per month and added Year, Month, Quarter and a Year Status flag that marks 2026 as a partial year.

Two formulas did most of the work. The first flags every row by type, and the second builds the State column, which holds a real state only for market rows:

```
=IF(D2="Geopolitical Zone","Zone",IF(OR(D2="National Average",D2="Market
Average"),"Benchmark",IF(D2="Kano State (1)","Unverified","Market")))

=IF(G2="Market",D2,"n/a")
```

3.4 Before and after cleaning

Aspect	Raw file	After cleaning
Rows	17,464	17,464 (none deleted; 15,576 are market rows used in calculations)
Location types	74 locations mixed together	66 Market, 5 Zone, 2 Benchmark, 1 Unverified
State column	Real states mixed with labels such as Market Average	State holds real states only, and n/a for all other rows
Region	Not in the file	Region column added
Date fields	price_date, year, month	A Date table of 236 months with quarter and a partial-year flag
Blank inflation values	888 per column	Unchanged on purpose
geo_id	Text IDs of different lengths	Unchanged; used as the key
Kano State (1)	Unexplained series	Marked Unverified and excluded, with the reason noted

3.5 Checks after cleaning

Check	Expected	Result
Locations with a unique geo_id	74	74
Market, Zone, Benchmark, Unverified	66, 5, 2, 1	66, 5, 2, 1
Distinct states among the markets	14	14
Blank latitude	1 (Market Average)	14
Dates in the Date table	236	236
Rows in the fact table	17,464	17,464
Months for each location	236	236
Fact IDs or dates missing from the dimension tables	0	0
Blank closing prices	0	0
Blank inflation values per column	888	888

3.6 Why the cleaning mattered

The 8 summary rows make up 1,888 of the 17,464 rows (10.8 percent). In this dataset, leaving them in would have changed the headline averages only slightly: the August 2026 average rice price would be about ₦4,718 instead of ₦4,724. But the results would still have been wrong in ways that matter. The market count would read 74 instead of 66, labels such as Market Average and Geopolitical Zone would appear as states in rankings, a row with no coordinates would break any map, and an unexplained series (Kano State (1)) would be counted as a real market. Cleaning therefore protected the counts, the rankings and the credibility of the dashboard more than the headline averages.

I also decided not to do certain things. I did not delete extreme inflation values, because they are real observations (I used the median and set the chart axis by hand instead). I did not fill in the blank inflation months. I did not merge or relabel Kano State (1) without evidence.

4. Data Modelling and Calculations

Once the data was clean, I organized it into a model and wrote the calculations that feed the dashboard.

4.1 Model design

I split the single sheet into a star schema: small dimension tables that describe places, dates and foods, and fact tables that hold the numbers.

Table	Rows	Role
Location	74	Dimension: state, LGA, market, region, coordinates, location_type
Date	236	Dimension: one row per month with year, month, quarter, and a flag for the partial year 2026
Commodity	8	Dimension: food name and category (staple or animal protein)
Prices	17,464	Fact (wide): one column per food, used for the food price index
PricesLong	139,712	Fact (long): one row per market, month and food

I created the long table because a wide table would have needed every food calculation to be written eight times. With one row per food, a single calculation works for every food and one slicer switches between them.

4.2 Building the tables in Excel

I worked only on copies of the raw data. For the Location table I removed duplicates on geo_id (74 rows), then used formulas to create location_type, State and Region. The State column holds a real state only for market rows and shows n/a otherwise, so summary labels never appear as states. For the Date table I generated 236 monthly dates with the EDATE function and added Year, Month Number, Month Name, Quarter and a Year Status column. For the Prices table I removed the seven descriptive columns that now live in the dimension tables, leaving 36 columns. After each table I ran checks, for example 74 unique location IDs, 236 dates, 17,464 fact rows, and 236 months for each of the 74 locations.

4.3 Building the model in Power BI

I loaded the tables, checked the data types, and created the Commodity table. I then built PricesLong in Power Query: I referenced the Prices table, removed the food index columns, unpivoted all other columns, split the column names at the first underscore into a type (c, h, l, inflation) and a food name, and pivoted the type back into columns named Close, High, Low and Inflation. The result had 139,712 rows, which matched the expected 17,464 times 8. I then connected the tables with five one-to-many relationships, marked Date as the date table, sorted month names by month number, and created a separate Measures table to hold all calculations.

4.4 Measures and design principles

I wrote 10 measures in DAX. Three principles guided them:

1. **Market rows only.** Every measure filter's location_type to Market so summary rows never distort the numbers.
2. **One food at a time.** Food measures return blank unless exactly one food is selected, because averaging rice, eggs and gari together gives a meaningless number when their units differ.
3. **Medians for inflation.** A few extreme months would pull an average well above the typical value.

The measures cover the number of markets, the average and latest food price index, median food inflation, the average and latest price of a food, its price in January 2007, the multiple of that starting price, median inflation per food, and the average and latest monthly high and low prices.

4.5 Validation

I checked the measures against values calculated independently from the raw file.

Check	Expected	Result
Markets	66	66
Latest Food Index (Aug 2026)	4.35	4.35
Latest Food Inflation, median	3.4	3.41
Rice, latest price	about 4,724	4723.65
Rice, price in Jan 2007	about 272	272
Rice, price multiple	about 17.4	17.37

5. The Dashboard

The dashboard has five pages. Each answer one question.

Page 1, Overview: what is happening to food prices overall? Year and Region slicers; cards for the number of markets, the latest food price index and the latest median food inflation; a line chart of the food price index over time; and a bar chart of the index by state.

Page 2, Prices: how much has each food cost over time? A food slicer set to select one food at a time; cards for the latest price, the January 2007 price and the multiple of that starting price; a line chart of the price over time; and a column chart of the average price by year.

Page 3, Inflation: how fast are prices rising, and for which foods? A line chart of median inflation over time and a bar chart comparing median inflation across the eight foods. I switched off the food slicer's effect on the bar chart so that it shows all foods.

Page 4, Geography: where are prices highest and lowest? A bar chart of average price by state, a column chart by region, and a table of markets sorted by price. **(NOT REPRESENTED YET)**

Page 5, About the Data: a notes page stating the coverage, the unit's issue, the exclusion of summary rows, and the other limitations, so a reader does not over-interpret the charts. **(NOT REPRESENTED YET)**

[Insert a screenshot of each page here, with a one-line caption.]

6. Findings

- **Prices rose sharply.** The average rice price across markets rose from about ₦272 in January 2007 to about ₦4,724 in August 2026, roughly 17.4 times. Using the National Average benchmark, the increase over the same period ranged from about 11 times for milk to about 22 times for rice and fish.
- **The sharpest rise was recent.** The food price index (National Average row) was 1.82 in 2023, 3.56 in 2024 and 4.06 in 2025. The 2026 average to August is 3.93.
- **Inflation is moderating but still positive.** The median year-on-year food inflation across markets in August 2026 was 3.4 percent. For rice, the median across all months was about 13.1 percent.
- **Prices differ by state.** In 2025 the average rice price was about ₦6,002 in Abia and about ₦4,023 in Sokoto, a gap of about 49 percent.
- **Some foods are far more volatile.** Large month-to-month jumps are common for fish, yam, beans and milk but absent for rice, gari, eggs and beef, and almost all the fish jumps occur in Borno and Yobe.

7. Challenges and How I Solved Them

Challenge	Cause	Solution
A syntax error on the Date table in DAX	Date is also the name of a DAX function	Wrote the table name in quotes, as 'Date'[Date]
The Pivot Column button was not visible	It sits in the Transform tab, Any Column group, of the Power Query Editor, and needs exactly one column selected	Selected only the type column
Bar chart wells confused me	Bar charts name wells by direction	Categories go in the Y-axis well and numbers in the X-axis well
Map error (location and latitude/longitude conflict)	The Location well and the coordinate wells were used together	Planned to use coordinates only; left the map for future work
Charts blank until a food was chosen	The measure is designed to return blank unless one food is selected	Set the slicer to single select with a default food

8. Limitations

- The North East holds 82 percent of the markets, so the results cannot be treated as national.
- The price units are not documented.
- Prices are nominal, so comparisons across many years should use multiples or indexes.

- The Region column is my own classification and should be verified, particularly the placement of Kaduna in the North West.
- Kano State (1) is unverified and excluded.
- 2026 contains only eight months, so it is not comparable with full years.
- The file does not define the monthly high and low; I interpreted them as the highest and lowest price seen in a market that month.

9. Conclusion and Recommendations

The project showed that most of the effort in a good dashboard lies in understanding and preparing the data. Identifying the summary rows, deciding how to treat Kano State (1), and reshaping the data into a clean model prevented misleading results, and the validation checks gave me confidence in the numbers. The dashboard shows that food prices have risen steeply since 2007, that the biggest jump came after 2023, and that prices vary widely between states.

I recommend the following next steps:

1. Confirm the price units and the meaning of Kano State (1), high and low with the data source.
2. Add a market map once the coordinates are verified.
3. Add a price index chart (January 2007 equals 100) to compare all foods on one scale.
4. Add an inflation heatmap by year and month, a volatility page based on the monthly price range, and a seasonality chart.
5. Adjust prices for inflation, or compare them with wages, to measure affordability.
6. Seek data from more southern markets to reduce the North East skew.

Acknowledgements and Use of Tools

[Check your institution's rules on AI assistance. If it is allowed, state here how you used it, for example: I used an AI assistant to explain data-cleaning approaches and to check my DAX formulas. All building, checking and interpretation of results were my own work. Also credit the data source.]

Appendix: Example Measures

```
Markets =
CALCULATE(DISTINCTCOUNT(Location[geo_id]), Location[location_type] = "Market")

Avg Price =
IF(
    HASONEVALUE(Commodity[Name]),
    CALCULATE(AVERAGE(PricesLong[Close]), Location[location_type] = "Market")
)

Latest Price =
VAR LastDt = MAX('Date'[Date])
RETURN
    CALCULATE([Avg Price], 'Date'[Date] = LastDt)

Price Multiple =
DIVIDE([Latest Price], [Base Price (Jan 2007)])
```